I checked your uploaded Python file, employe salary.py. I'll prepare the complete activity record based on your actual code, using simple language suitable for a B.Tech 1st-year CSE Data Science student.

Your program calculates an employee's gross salary, applies a deduction based on the gross salary, and displays the final net salary.

Employee Salary Calculation System

1. Problem statement

To develop a Python program that calculates an employee's salary based on basic salary, overtime pay, and incentives. The program calculates house rent allowance (HRA) and additional allowance, determines the deduction rate based on gross salary, and displays the gross salary, deduction, and net salary.

2. Objectives

1. To calculate HRA and allowance using the basic salary.
2. To calculate the employee's gross salary.
3. To apply a deduction rate based on the gross salary.
4. To calculate the final net salary.
5. To display the employee's salary statement.
6. To understand variables, arithmetic operations, and if-else statements in Python.

3. Assumptions

- The employee's name is entered as text.
- Basic salary, overtime pay, and incentive are entered as numerical values.
- HRA is calculated as 20% of the basic salary.
- Allowance is calculated as 10% of the basic salary.
- A 5% deduction is applied when gross salary is ₹30,000 or less.
- A 10% deduction is applied when gross salary is greater than ₹30,000.
- The program uses the deduction rates specified in the code; these are assumed rules for this exercise, not necessarily actual payroll or tax rules.

4. Algorithm

1. Start the program.
2. Enter the employee's name.
3. Enter the basic salary, overtime pay, and incentive.
4. Calculate HRA as 20% of the basic salary.
5. Calculate the allowance as 10% of the basic salary.
6. Calculate the gross salary by adding the basic salary, HRA, allowance, overtime pay, and incentive.

7. If the gross salary is ₹30,000 or less, set the deduction rate to 5%.
8. Otherwise, set the deduction rate to 10%.
9. Calculate the deduction amount.
10. Calculate the net salary by subtracting the deduction from the gross salary.
11. Display the employee's name, gross salary, deduction, and net salary.
12. Stop the program.

5. Flowchart

Start

Input employee name, basic salary, overtime, and incentive

Calculate HRA (20%) and allowance (10%)

Calculate gross salary

Is gross salary ≤ ₹30,000?

Yes

Deduction rate = 5%

No

Deduction rate = 10%

Calculate deduction and net salary

Display salary statement

End

6. Pseudocode

```
BEGIN
   INPUT name
   INPUT basic
   INPUT overtime
   INPUT incentive

   hra = basic * 0.20
   allowance = basic * 0.10

   gross = basic + hra + allowance
         + overtime + incentive
```

```
    IF gross <= 30000 THEN
        rate = 0.05
    ELSE
        rate = 0.10
    END IF

    deduction = gross * rate
    net = gross - deduction

    DISPLAY name
    DISPLAY gross salary
    DISPLAY deduction
    DISPLAY net salary
END
```

7. Python source code

```
name = input("Enter employee name: ")
basic = float(input("Enter basic salary: "))
overtime = float(input("Enter overtime pay: "))
incentive = float(input("Enter incentive: "))

hra = basic * 0.20
allowance = basic * 0.10

gross = basic + hra + allowance + overtime + incentive

if gross <= 30000:
    rate = 0.05
else:
    rate = 0.10

deduction = gross * rate
```

```
net = gross - deduction


print("\n--- SALARY STATEMENT ---")
print("Employee:", name)
print("Gross salary: ₹", round(gross, 2))
print("Deduction: ₹", round(deduction, 2))
print("Net salary: ₹", round(net, 2))
```

8. Test cases

Test	Basic salary	Overtime	Incentive	Expected deduction rate
1	₹20,000	₹1,000	₹500	5%
2	₹25,000	₹2,000	₹1,000	10%
3	₹25,000	₹0	₹0	5%
4	₹30,000	₹0	₹0	10%
5	₹10,000	₹500	₹500	5%

Important: Test 4 has a gross salary of ₹39,000 because HRA and allowance are added. Therefore, the deduction rate is 10%, even though the basic salary is ₹30,000.

9. Sample output screenshots

The following are sample console previews, not actual screenshots captured from a running Python interpreter.

Test 1 — Gross salary below ₹30,000

```
Enter employee name: Ravi
Enter basic salary: 20000
Enter overtime pay: 1000
Enter incentive: 500

--- SALARY STATEMENT ---
Employee: Ravi
Gross salary: ₹ 27500.0
Deduction: ₹ 1375.0
Net salary: ₹ 26125.0
```

Copy sample output

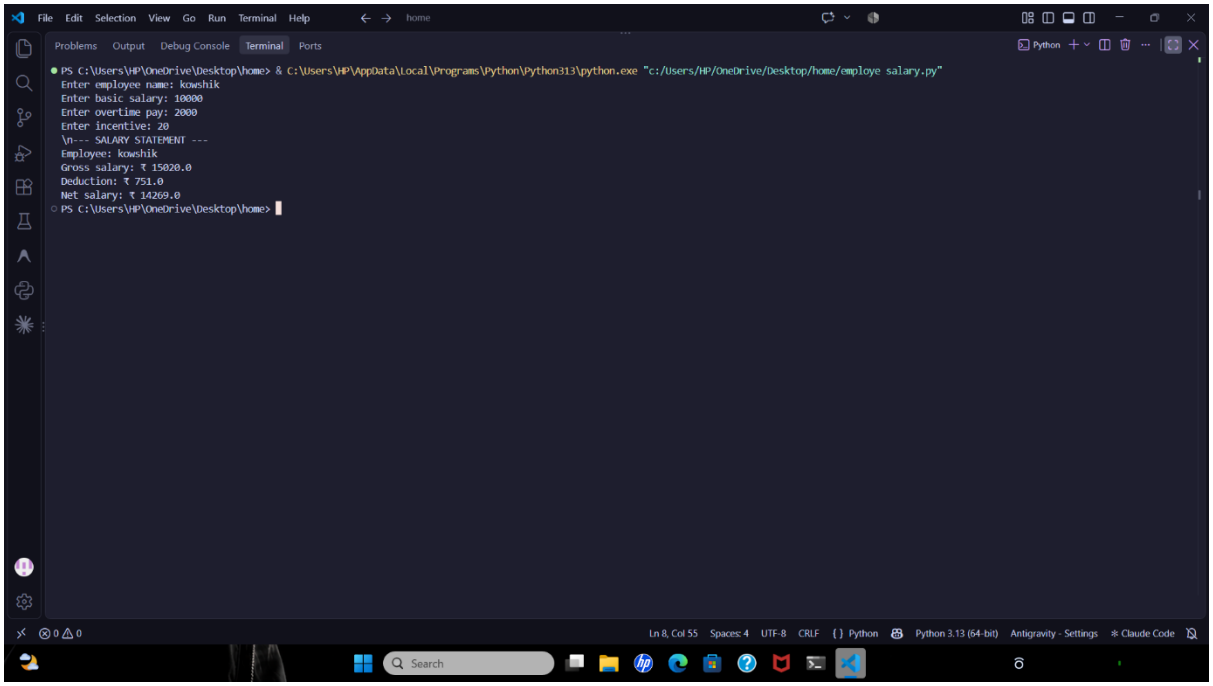
Test 2 — Gross salary above ₹30,000

```
Enter employee name: Sita
Enter basic salary: 25000
Enter overtime pay: 2000
Enter incentive: 1000
```

--- SALARY STATEMENT ---
Employee: Sita
Gross salary: ₹ 35500.0
Deduction: ₹ 3550.0
Net salary: ₹ 31950.0

Copy sample output

To get actual screenshots for your record, run the program in Python IDLE or VS Code and capture the console output.



10. Conclusion

The Employee Salary Calculation System was successfully developed using Python. The program calculates HRA, allowance, gross salary, deduction, and net salary based on the given inputs. It uses an if-else statement to select the deduction rate according to the gross salary. This program helps in understanding variables, arithmetic operations, conditional statements, and input/output functions in Python.